

Student Information System

User Manual

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This manual explains how to use the Student Information System, including student records, course management, grades, GPA calculation, and transcript generation.

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Introduction

The Student Information System is a console-based C++ application used to manage university student data. It allows the user to add, view, search, update, and delete student records. It also supports course management, course registration, grade entry, GPA calculation, and transcript generation.

The program stores data in CSV files, so the saved information can be loaded automatically the next time the program runs.

System Requirements

Data Files Used by the Program

The program automatically creates and uses the following CSV files in the same folder as the executable:

File Name	Purpose
students.csv	Stores student personal data, academic data, and registered courses.
courses.csv	Stores course code, course title, and credit hours.
grades.csv	Stores student grades, totals, letter grades, and grade points.

Starting the Program

When the program starts, it loads saved data from the CSV files. Then it displays the main menu:

```
=====
|| STUDENT INFORMATION SYSTEM ||
=====
1. Student Management
2. Courses Management
3. Grades Management
4. Exit
Enter your choice:
```

Type the number of the option you want, then press **Enter**.

Main Menu Options

Option	Description
1	Opens Student Management. Use this to add, list, search, update, or delete students.
2	Opens Courses Management. Use this to add, view, update, delete, or register courses to students.
3	Opens Grades Management. Use this to enter grades, view grades, calculate GPA, or generate transcripts.
4	Exits the program.

Student Management

Student Management is used to control all student records.

```
=== Student Management ===
1. Add New Student
2. List All Students
3. Search Students
4. Update Students
5. Delete student
6. Back to Main Menu
```

Add New Student

Choose option **1** from the Student Management menu. The program will ask for the following information:

Field	Required Format
Student Name	Exactly two names, using letters only. Example: Omar Ali .
National ID	Must be 14 digits, must not start with 0, and must be unique.
Gender	Must be M, m, F, or f.
Date of Birth	Must be in DD/MM/YYYY format. Student age must be at least 17.
Phone Number	Must be 11 digits and start with 01.
Program	Must be CSE, CCE, or MCT.
Academic Level	Must be from 1 to 4.

After entering valid data, the system creates a Student ID automatically. The ID format is:

YYPXXXX

For example, 26P0001 means the student was added in year 2026 and is student number 0001.

List All Students

Choose option **2** to show all students. The program asks how to sort them:

- By Student ID
- By Name from A to Z
- By GPA from highest to lowest

The list displays the Student ID, name, program, level, and GPA.

Search Students

Choose option **3**. You can search using:

1. Student ID
2. Student Name
3. National ID

When a match is found, the program displays the full student details, including registered courses.

Update Student

Choose option **4**. Enter the Student ID of the student you want to update. If the student exists, you can update:

- Phone number
- Program
- Academic level

After updating, the change is saved automatically to **students.csv**.

Delete Student

Choose option **5**. Enter the Student ID. The program asks for confirmation using Y or N. If you confirm, the student is removed and the file is updated.

Important

Deleting a student removes the student record from the student list. Use this option carefully.

Course Management

Course Management is used to create and control course records, and to register students in courses.

```
=== Course Management ===
1. Add New Course
2. View All Courses
3. Update Course
4. Delete Course
5. Register Course to Student
6. Back to Main Menu
```

Add New Course

Choose option **1**. The program asks for:

Field	Required Format
Course Code	Must be 3 capital letters followed by 3 digits. Example: CSE141.
Course Title	Cannot be empty.
Credit Hours	Must be from 1 to 4.

The course code must be unique. If the course code already exists, the program will ask for another one.

View All Courses

Choose option **2**. The program displays a table containing:

- Course code
- Course title
- Credit hours

Update Course

Choose option **3**. Enter the course code. If the course is found, you can update:

- Course title
- Credit hours

Delete Course

Choose option **4**. Enter the course code. The program asks for confirmation before deleting the course.

Register Course to Student

Choose option **5**. The program asks for:

1. Student ID
2. Course Code

The program checks that the student exists, the course exists, and the student is not already registered in the same course. A student can register in a maximum of 10 courses.

Grades Management

Grades Management is used to enter student grades, view grades, calculate GPA, and generate transcripts.

```
=== Grades Management ===
1. Enter Student Grades
2. View Student Grades
3. Calculate GPA
4. Generate Transcript
5. Back to Main Menu
```

Enter Student Grades

Choose option **1**. The program asks for:

Field	Required Format
Student ID	Must belong to an existing student.
Course Code	Must belong to an existing course.
Registration Check	The student must already be registered in the course.
Midterm Grade	Must be between 0 and 40.

Final Grade Must be between 0 and 60.

The total grade is calculated as:

$$\text{Total} = \text{Midterm Grade} + \text{Final Grade}$$

The program then assigns a letter grade and grade points.

Letter Grade Rules

Total Grade	Letter Grade	Grade Points
90 to 100	A	4.0
80 to 89	B	3.0
70 to 79	C	2.0
60 to 69	D	1.0
Below 60	F	0.0

View Student Grades

Choose option **2**. Enter the Student ID. The program displays all grades recorded for that student, including course code, midterm, final, total, and letter grade.

Calculate GPA

Choose option **3**. Enter the Student ID. The GPA is calculated using grade points and course credit hours:

$$\text{GPA} = \text{Total Weighted Points} / \text{Total Credit Hours}$$

Where:

- Total Weighted Points = sum of grade points multiplied by credit hours.
- Total Credit Hours = sum of credit hours for courses with recorded grades.

After calculation, the GPA is saved inside the student record.

Generate Transcript

Choose option **4**. Enter the Student ID. The transcript displays:

- Student ID
- Student name
- Program and academic level
- Course code and title
- Credit hours
- Letter grade
- Grade points
- Total credit hours
- Cumulative GPA

Suggested Correct Workflow

For best results, use the system in this order:

1. Add students first.
2. Add courses.
3. Register students in courses.
4. Enter grades for registered courses.
5. Calculate GPA.
6. Generate transcript.

Example Workflow

Add student → Add course → Register course to student → Enter grades → Calculate GPA → Generate transcript.

Troubleshooting

Problem	Possible Solution
Program says “Student not found”	Make sure the Student ID is typed exactly as shown, for example 26P0001.
Program says “Course not found”	Make sure the course was added before registering or entering grades.
Cannot enter grades	Check that the student exists, the course exists, and the student is already registered in that course.
GPA shows no grades available	Enter grades first, then calculate GPA.
Saved data does not appear	Make sure the CSV files are in the same folder as the program executable.
Invalid input message appears	Check the required format for that field and enter the value again.

Notes for Users

- Always enter menu choices as numbers.
- Use capital letters for course codes and program names.
- Do not manually edit CSV files unless you understand the file format.
- Keep backup copies of `students.csv`, `courses.csv`, and `grades.csv` if the data is important.
- Exit the program normally from the main menu after finishing your work.

Conclusion

This Student Information System provides a simple way to manage student records, courses, grades, GPA, and transcripts. By following the correct workflow and input formats, the user can safely store and retrieve student academic information through the console interface.